

Topic C12

Experimental techniques and chemical analysis

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

c12.1	Apparatus and measurement
c12.2	Separation and purification
c12.3	Chromatography
c12.4	Tests for gases
c12.5	Tests for ions
c12.6	Experimental planning and evaluation

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Measurement and apparatus

Choose apparatus according to precision and purpose. A burette measures variable accurate volumes; a volumetric pipette transfers one fixed accurate volume; a measuring cylinder is less precise; a gas syringe measures gas volume. Read liquid levels at eye level from the bottom of a concave meniscus and record burette readings to the appropriate precision.

Examiner insight

A beaker is for containing or approximate volumes, not accurate measurement.

2. Separation and purification

Filtration separates an insoluble solid from a liquid. Crystallisation obtains a soluble solid by concentrating a solution and cooling it. Simple distillation obtains solvent or separates liquids with widely different boiling points; fractional distillation separates miscible liquids with closer boiling points. Purity is checked by a sharp melting or boiling point.

Examiner insight

Choose the method from the physical properties: solubility, particle size and boiling point.

3. Chromatography

Paper chromatography separates soluble substances by different relative attractions to stationary paper and mobile solvent. Draw the baseline in pencil, keep it above the solvent and mark the solvent front. A pure substance gives one spot in one solvent. Compare R_f values only under identical conditions.

$R_f = \text{distance moved by substance} / \text{distance moved by solvent front}$

Worked example

A spot moves 4.2 cm and solvent 6.0 cm: $R_f = 0.70$.

Examiner insight

A colourless sample needs a locating agent; an R_f value alone does not prove identity.

4. Gas tests

Hydrogen gives a squeaky pop with a lighted splint. Oxygen relights a glowing splint. Carbon dioxide turns limewater milky. Ammonia turns damp red litmus blue. Chlorine bleaches damp litmus paper after first turning it red. Tests use small samples and appropriate safety precautions.

Examiner insight

Give both the procedure and the positive observation.

5. Tests for aqueous ions

Flame tests: Li^+ crimson, Na^+ yellow, K^+ lilac, Ca^{2+} orange-red and Cu^{2+} blue-green. With aqueous sodium hydroxide: Cu^{2+} gives a blue precipitate, Fe^{2+} green, Fe^{3+} red-brown, and NH_4^+ releases ammonia on warming. Halides use acidified silver nitrate; sulfate uses acidified barium nitrate; carbonate releases carbon dioxide with acid.

Examiner insight

Acidifying removes interfering ions; use the specified reagent and observation.

6. Planning, errors and evaluation

A valid plan identifies independent, dependent and control variables; gives a reproducible method; states measurements, range, repeats, hazards and precautions. Random error is reduced by repeats and means; systematic error shifts results consistently and needs recalibration or method change. An anomalous result is identified from the pattern, not simply because it is inconvenient.

Examiner insight

Improvements must address a named limitation directly, such as using a gas syringe instead of counting bubbles.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic C12

Experimental techniques and chemical analysis

Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which apparatus transfers 25.0 cm³ most accurately? [1]
 - A beaker
 - B pipette
 - C conical flask
 - D test tube

2. Which method separates an insoluble solid from a liquid? [1]
 - A filtration
 - B distillation
 - C chromatography
 - D titration

3. Which method obtains pure water from salt solution? [1]
 - A filtration
 - B simple distillation
 - C crystallisation only
 - D paper chromatography

4. Why is a chromatography baseline drawn in pencil? [1]
 - A Graphite dissolves readily.
 - B Graphite does not dissolve in the solvent.
 - C Ink is colourless.
 - D Pencil increases R_f.

5. A spot moves 3 cm and the solvent front 5 cm. What is R_f? [1]
 - A 0.6
 - B 1.7
 - C 2.0
 - D 8.0

6. Which gas relights a glowing splint? [1]
 - A hydrogen
 - B oxygen
 - C carbon dioxide
 - D ammonia

7. Which gas turns limewater milky? [1]
 - A chlorine
 - B hydrogen
 - C carbon dioxide
 - D oxygen

8. A yellow flame indicates [1]
- A lithium
 - B sodium
 - C potassium
 - D copper
9. A blue precipitate with sodium hydroxide indicates [1]
- A Cu^{2+}
 - B Fe^{2+}
 - C Fe^{3+}
 - D NH_4^+
10. How is sulfate identified? [1]
- A acidified barium nitrate gives a white precipitate
 - B silver nitrate gives a cream precipitate
 - C sodium hydroxide gives a blue precipitate
 - D litmus is bleached

Section B · Structured questions

11 A student titrates an acid with an alkali.

(a) State the purpose of a pipette. [2]

(b) Explain how a burette reading is taken. [2]

(c) Explain why a conical flask is used. [2]

(d) Describe how concordant results improve reliability. [2]

[Total: 8]

12 A mixture contains sand, sodium chloride and water.

(a) Describe how to obtain dry sand. [2]

(b) Describe how to obtain sodium chloride crystals. [3]

(c) Describe how to obtain pure water. [2]

(d) State one purity test for the water. [1]

[Total: 8]

13 A dye spot moves 4.8 cm while the solvent front moves 8.0 cm.

(a) Calculate the R_f value. [2]

(b) State why the baseline must remain above the solvent. [2]

(c) Explain what two spots from one sample indicate. [2]

(d) State two conditions needed for valid comparison. [2]

[Total: 8]

14 An unknown solution may contain ammonium, copper(II), sulfate and chloride ions.

(a) Describe the test for ammonium ions. [2]

(b) Describe the test for copper(II) ions. [2]

(c) Describe the test for sulfate ions. [2]

(d) Describe the test for chloride ions. [2]

[Total: 8]

15 Plan an investigation of temperature and reaction rate.

(a) Identify independent and dependent variables. [2]

(b) State two control variables. [2]

(c) Describe how results are made reliable. [2]

(d) State one hazard, precaution and linked improvement. [2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

1 B. A volumetric pipette transfers a fixed accurate volume.

2 A. The solid remains in the filter.

3 B. Water vaporises and is condensed.

4 B. It does not produce extra chromatogram spots.

5 A. $3/5 = 0.6$.

6 B. This is the test for oxygen.

7 C. A calcium carbonate precipitate forms.

8 B. Sodium ions produce a yellow flame.

9 A. Copper(II) hydroxide is blue.

10 A. Insoluble barium sulfate forms.

Solutions · Structured questions

11. 11(a) It transfers a fixed accurate volume [2]. (b) Read at eye level from the bottom of the meniscus and record initial and final readings [2]. (c) It allows swirling without loss while retaining all moles [2]. (d) Repeat until close titres are obtained and calculate a mean, reducing random error [2].

12. 12(a) Filter, wash the residue and dry it [2]. (b) Evaporate the filtrate to saturation, cool, filter and dry crystals [3]. (c) Use simple distillation and collect the condensate [2]. (d) A sharp boiling point of 100 °C at standard pressure [1].

13. 13(a) $4.8/8.0 = 0.60$ [2]. (b) It prevents the sample dissolving directly into the solvent reservoir [2]. (c) At least two soluble substances are present, so it is a mixture [2]. (d) Same solvent, paper, temperature or run conditions [any 2].

14. 14(a) Add sodium hydroxide and warm; ammonia turns damp red litmus blue [2]. (b) Add sodium hydroxide; a blue precipitate forms [2]. (c) Acidify and add barium nitrate; a white precipitate forms [2]. (d) Acidify with nitric acid and add silver nitrate; a white precipitate forms [2].

15. 15(a) Temperature; time or rate from gas volume, mass loss or colour change [2]. (b) Same concentrations, volumes, particle size or apparatus [any 2]. (c) Repeat each temperature, identify anomalies and calculate means [2]. (d) Valid hazard with precaution plus a method improvement linked to precision [2].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.